

CSE 5210

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Problem Set 1

Problem 1:

$X = 1$ when $C = 0$ & $D = 1$

$Y = 1$ when $(A,B,C) \geq 5$

Truth Table:

A	B	C	D	X	Y
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0	0	0	0	0	0
---	---	---	---	---	---

0	0	0	1	1	0
---	---	---	---	---	---

0	0	1	0	0	0
---	---	---	---	---	---

0	0	1	1	0	0
---	---	---	---	---	---

0	1	0	0	0	0
---	---	---	---	---	---

0	1	0	1	1	0
---	---	---	---	---	---

0	1	1	0	0	0
---	---	---	---	---	---

0	1	1	1	0	0
---	---	---	---	---	---

1	0	0	0	0	0
---	---	---	---	---	---

1	0	0	1	1	0
---	---	---	---	---	---

1	0	1	0	0	1
---	---	---	---	---	---

1	0	1	1	0	1
---	---	---	---	---	---

1	1	0	0	0	1
---	---	---	---	---	---

1 1 0 1 | 1 1

1 1 1 0 | 0 1

1 1 1 1 | 0 1

K-map:

$X = 1$

AB = 00, CD = 01

AB = 01, CD = 01

AB = 10, CD = 01

AB = 11, CD = 01

CD

00 01 11 10

AB 00 0 1 0 0

01 0 1 0 0

11 0 1 0 0

10 0 1 0 0

$X = C' \text{ AND } D$

$Y = 1$

AB = 10, CD = 10

AB = 10, CD = 11

AB = 11, CD = 00

AB = 11, CD = 01

AB = 11, CD = 10

AB = 11, CD = 11

CD

00 01 11 10

AB 00 0 0 0 0

 01 0 0 0 0

 11 1 1 1 1

 10 0 0 1 1

$Y = A \text{ AND } B \text{ OR } A \text{ AND } C$

Problem 2:

Truth Table:

F = 1 when input between 9 and 12

A B C D | F

0 0 0 0 | 0

0 0 0 1 | 0

0 0 1 0 | 0

0 0 1 1 | 0

0 1 0 0 | 0

0101|0
0110|0
0111|0
1000|0
1001|1
1010|1
1011|1
1100|1
1101|0
1110|0
1111|0



